

Angular Manifold Routing: Fixed Geometric Routing from Embedding Angular Structure

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Abstract

We study whether the angular non-uniformity of normalized token embeddings can be exploited for routing computation in transformer-style architectures. We propose a fixed geometric routing method based on a Hopf-fibration-derived sector map, and evaluate it in two settings: a semantic proxy based on PPMI-SVD embeddings, and a small trainable language model.

On the semantic proxy, the effective routing footprint grows sublinearly with routing budget, following an empirical scaling law of approximately $K^{0.572}$, compared with $K^{1.0}$ for dense routing. This advantage persists through $K = 5000$. In a 2-layer trainable language model, fixed geometric routing replaces a learned top-1 gate with an 8% validation perplexity cost and no learned gate matrix, while retaining a concentrated routing footprint. A second-dataset replication on WikiText-2 closely matches the Penn Treebank result.

A key limitation is that in the trainable language model, the specific Hopf sector geometry is not distinguishable from randomly permuted fixed routing, suggesting that expert weights co-adapt to any fixed routing scaffold in this small-scale regime. The contribution of this work is therefore narrow: angular structure supports sparse fixed routing and transfers partially to trainable routing, but geometry-specific quality gains are not established in end-to-end language model training.

1 Introduction

Routing and sparsity are central themes in efficient transformer design. Mixture-of-experts (MoE) models and related sparse architectures improve scaling by activating only a subset of available compute paths for a given input (Shazeer et al., 2017; Fedus et al., 2022). Most such approaches rely on learned gating, which introduces additional parameters and training dynamics of its own.

This paper studies a different question: can some routing structure be obtained directly from the geometry of the representation space, without a learned routing network? The motivating observation is that normalized token embeddings are not uniformly distributed on the unit hypersphere. Recent work on compression has shown that this angular structure can be exploited for efficient representation (Google Research, 2026). Here we ask whether the same broader geometric fact

can support fixed routing.

Our approach uses a fixed routing map derived from a 4D Hopf fibration projection. The paper makes three scoped contributions:

1. On a semantic proxy, fixed geometric routing yields a sub-linear effective routing footprint over a broad range of routing budgets.
2. In a small trainable language model, fixed routing partially transfers, replacing a learned top-1 gate at modest perplexity cost.
3. In that trainable setting, the specific Hopf geometry does not outperform randomly permuted fixed routing, clarifying the scope of the result.

The intended claim is narrow. This work does *not* establish large-scale replacement of learned MoE routing in realistic systems. It presents an empirical efficiency result in a proxy setting and a toy-scale trainable language model.

2 Background

The paper is motivated by the observation that normalized language-model embeddings exhibit exploitable angular structure. Compression work such as TurboQuant treats this as a representation property to be transformed for scalar quantization (Google Research, 2026). Our focus is different: we ask whether a fixed geometric partition of the sphere can induce a non-uniform routing distribution that concentrates traffic into a subset of available sectors.

This framing leads naturally to a fixed-routing problem. If semantically related points cluster into preferred angular regions, then a sector map on the sphere should receive non-uniform traffic, potentially reducing the effective routing footprint without learning a routing matrix.

3 Method

3.1 Fixed geometric routing

Given an L2-normalized input vector $x \in \mathbb{R}^D$, we project to a 4D subspace using four fixed coordinates:

$$(a, b, c, d) = (x_i, x_j, x_k, x_l).$$

In the canonical configuration used here, the selected indices are (3, 65, 2, 21).

Table 1: Proxy routing scaling laws.

Routing	Scaling law	R^2
HOPF	$\hat{e} = 2.957 \cdot K^{0.572}$	0.963
PERMUTED	$\hat{e} = 1.664 \cdot K^{0.814}$	0.993
DENSE	$\hat{e} = K^{1.0}$	1.000

We then compute Hopf-base coordinates:

$$\rho_1 = \sqrt{a^2 + b^2}, \quad \rho_2 = \sqrt{c^2 + d^2}, \quad (1)$$

$$\chi_u = \frac{\rho_2^2}{\rho_1^2 + \rho_2^2}, \quad \delta = \text{atan2}(b, a) - \text{atan2}(d, c) \quad (2)$$

$$\alpha = \frac{1}{2} (\text{atan2}(b, a) + \text{atan2}(d, c)). \quad (3)$$

An optional transport correction is applied:

$$\alpha_t = \alpha + \frac{1}{2} \lambda \cos(2\chi) \delta,$$

where χ is derived from (ρ_1, ρ_2) and λ controls the correction strength. The coordinates are discretized into a product sector budget K , yielding a fixed routing sector.

3.2 Effective routing footprint

To measure routing concentration, we use an effective-buckets metric $\hat{e}$, which summarizes how many sectors are effectively active under the empirical routing distribution. Lower $\hat{e}$ relative to K indicates stronger concentration and lower effective routing footprint.

4 Proxy Experiments

4.1 Setup

We first evaluate the routing mechanism on a semantic proxy based on PPMI-SVD embeddings derived from standard distributional semantics pipelines (Mikolov et al., 2013; Pennington et al., 2014). This setting isolates routing geometry from end-to-end expert co-adaptation.

We compare the fixed Hopf routing map against dense routing and randomized controls across routing budgets ranging from small to large K .

4.2 Main result

Across the proxy experiments, the effective routing footprint scales approximately as

$$\hat{e} \approx 2.957 \cdot K^{0.572}$$

for the canonical Hopf routing configuration, compared with linear $K^{1.0}$ scaling for dense routing. At larger budgets, the observed advantage persists through $K = 5000$, with ratios in the range of approximately 2.6 to 2.8 $\times$.

The proxy result supports the core geometric claim of the paper: fixed angular routing can exploit non-uniform structure in normalized embeddings to produce a sparse routing footprint.

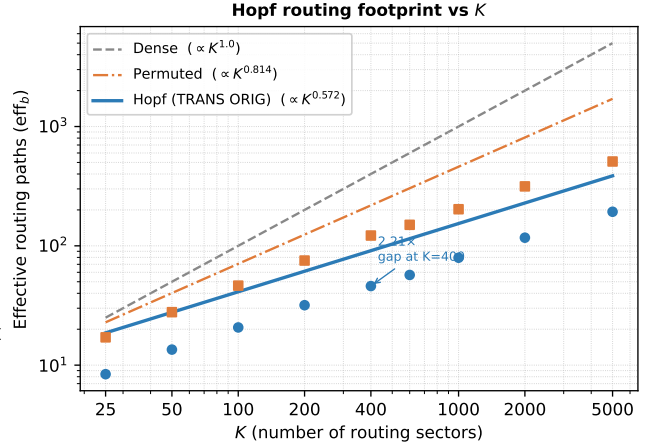


Figure 1: Effective routing footprint versus routing budget. Fixed Hopf routing shows sublinear growth relative to dense routing.

5 Trainable Language Model Experiments

5.1 Setup

We evaluate the method in a 2-layer transformer language model with hidden size 128, four attention heads, sequence length 32, vocabulary size 5000, and routing budget $K = 64$. Three conditions are compared:

- **BASELINE**: learned top-1 routing gate;
- **HOPF-ROUTED**: fixed geometric routing with no learned gate matrix;
- **PERMUTED**: randomly permuted fixed routing sectors.

Training uses 4000 steps and two datasets: Penn Treebank (Marcus et al., 1993) and WikiText-2 (Merity et al., 2016).

5.2 Penn Treebank and WikiText-2

On Penn Treebank, fixed geometric routing reaches within 8% of the learned-gating baseline in validation perplexity while retaining a concentrated routing footprint and using no learned gate matrix. A second-dataset replication on WikiText-2 yields a closely matching HOPF/BASELINE ratio under the same training setup.

5.3 Null result: geometry-specific quality gain

A key result of the paper is that HOPF-ROUTED and PERMUTED are effectively indistinguishable in the trainable language model. This suggests that in the small trainable regime studied here, expert weights co-adapt to the fixed routing scaffold, and the specific Hopf geometry does not provide additional quality advantage.

This sharpens the claim of the paper: the sparse fixed-routing mechanism transfers, but the specific geometric scaffold does not show a trainable-LM quality advantage over random fixed routing.

Table 2: Cross-dataset summary for the trainable LM experiments.

Metric	PTB	WikiText-2
HOPF/BASELINE PPL ratio	1.081	1.081
HOPF vs. PERMUTED $ \Delta\text{ppl} $	0.13	0.03
HOPF vs. DENSE effective-path ratio	$1.39\times$	$1.56\times$

6 Discussion

The empirical picture is therefore mixed but informative. On the one hand, the proxy experiments show a strong sublinear routing footprint derived from embedding geometry alone. On the other hand, end-to-end language-model training absorbs the geometry-specific advantage, leaving a narrower result: fixed routing can replace learned gating at moderate cost in a toy-scale setting, but the particular geometric scaffold is not uniquely beneficial in that trainable regime.

This positions the contribution as a narrow efficiency result rather than a broad architecture claim. The proxy experiments establish a geometry-derived routing footprint. The trainable experiments establish partial transfer of that footprint and simultaneously identify a null result: the specific Hopf sector geometry does not appear to matter once expert weights co-adapt.

7 Limitations

This work has several important limitations.

1. The strongest scaling-law results are established on a semantic proxy rather than large end-to-end language models.
2. The trainable experiments use a 2-layer model and should not be interpreted as evidence for large-scale replacement of learned routing.
3. The Hopf geometry does not outperform randomly permuted fixed routing in the trainable language model setting.
4. The experiments do not address attention-side routing.
5. All experiments were conducted by a single researcher.

8 Conclusion

We presented a fixed geometric routing method based on a Hopf-fibration-derived sector map and evaluated it on both a semantic proxy and a small trainable language model. The main empirical result is that angular structure in normalized embeddings supports a sublinear effective routing footprint and can partially transfer to a trainable routing setting without a learned gate matrix. The main limitation is that the specific geometry does not provide an observable quality advantage over random fixed routing in the trainable regime studied here.

The contribution is therefore narrow but concrete: embedding angular structure appears relevant not only for compression but also for routing computation.

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